Customer income classification and segmentation

Census-Income

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1 Executive summary

The retail client wants to identify customers earning more than $50K for targeted marketing. On average, mailers cost roughly $2 per recipient, and a converted high-income customer is worth meaningfully more than a converted low-income one, so missing a real high earner is more expensive than mailing one who turns out to be low income. I built the evaluation around that asymmetry, using PR-AUC and recall at fixed mailer budgets rather than accuracy.

The final classifier is a calibrated XGBoost model with sampling-weight-aware evaluation. It reaches PR-AUC of 0.655 and captures 75.1% of high-income customers within the top 10% of scored individuals, compared to 70.7% for a logistic regression baseline. On the test set, that gap translates to roughly 600 additional correctly identified high earners at the same mailer spend.

For segmentation I used a Gaussian Mixture Model on a TruncatedSVD projection of the same features, producing four segments. One segment (32.5% of the population) is the prime marketing target: full-year working adults with a 13% high-income rate. One segment (39.4%) is non-participants and should be excluded entirely. The remaining two are an immigrant working-household segment and a younger transitional-worker segment, each with its own product fit.

2 Data and business framing

The dataset is the Census-Income (KDD) extract from the 1994 and 1995 U.S. Current Population Surveys, with about 200,000 rows, 40 demographic and employment variables, a sampling weight per record, and a binary income label. The positive class is roughly 6% of the rows, and the imbalance is similar after applying the survey weights. The weight reflects how many people in the general population each row represents under the stratified sampling design.

Class imbalance at this level makes accuracy irrelevant as a metric. A model that predicts everyone as low-income gets 94% accuracy and zero business value. Three metrics drove the work instead. PR-AUC, because the positive class is rare and PR curves are more informative than ROC under imbalance [1]. Recall at top-k percent, because the client operates on a fixed mailer budget and wants to know what fraction of true high earners they capture for that spend. Brier score, because if the client ever wants to set a probability threshold rather than rank-cut, the scores need to be accurate.

3 Exploratory data analysis and preprocessing

The first useful finding was that the dataset has two different kinds of missingness. Variables like `reason for unemployment`, `enroll in edu inst last wk`, and the veteran questionnaire field are 90 to 100% "Not in universe." That is structural: most people have never been unemployed or enrolled in school in the last week, so the question does not apply. The four migration-code columns, by contrast, are about 50% literal "?", meaning genuinely unknown. I treated structural NIU as a valid category and replaced "?" with an explicit "Missing" token so the model could learn from the pattern of missingness itself.

The categorical EDA pointed at a small number of features doing most of the predictive work. Education shows a clean income staircase from children at near zero through to professional and doctorate holders at about 53%. Marital status compresses to three groups with a roughly 8x gap between married and never-married. Occupation produces a 13x gap between executive or managerial roles and service work. Sex shows the 1994 wage gap directly, with male high-income rate at about 10% versus 2.6% for female.

Race and country-of-birth patterns reflect the 1994 demographic structure of the sampled population and should not be read as present-day statements. Income distributions across these groups have shifted meaningfully since then, and *any deployed model would need revalidation on modern data before being used for decisions that affect those groups.*

The financial variables (capital gains, capital losses, dividends, wage per hour) are extreme spike-at-zero distributions. Out of 200,000 records, around 190,000 have zero capital gains, zero dividends, and zero capital losses. Raw values are unusable: a single $99,999 outlier dominates. Even after a $\log(1 + x)$ transform the zero spike is still the dominant mass, but the tail is now compressed enough to be useful.

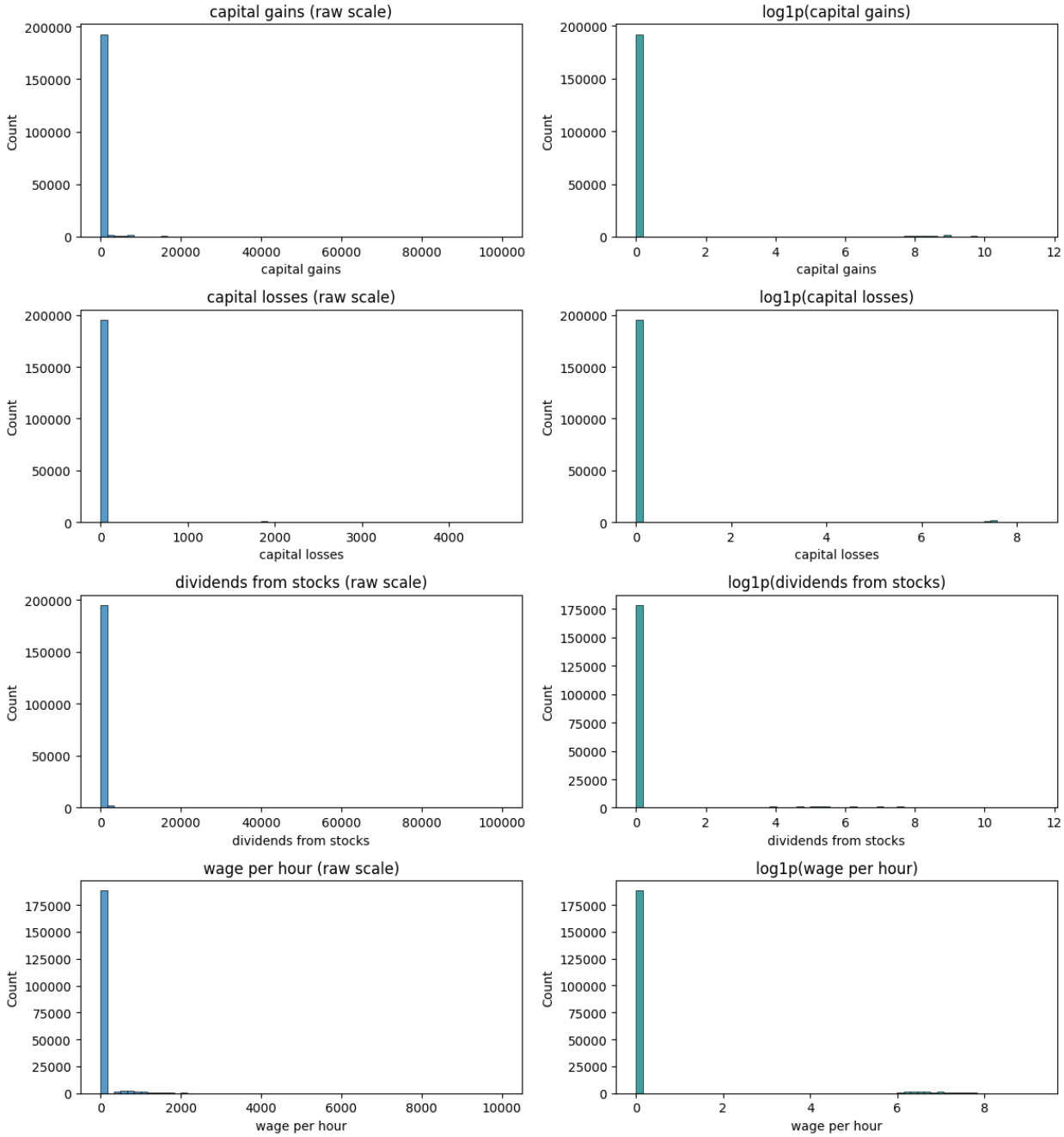

Figure 1: Raw (left column) versus $\log(1 + x)$ (right column) distributions for the four financial variables. The raw scale is unusable: a handful of extreme values dominate the axis and compress the bulk of the distribution into a single bar at zero. After the log transform the zero spike is still the dominant mass, but the nonzero tail is now resolvable, which is what the model needs to pick up the investment-income signal.

I kept both the log-transformed value and a binary presence flag, because the EDA showed that whether a person had any investment income at all carried more signal than how much.

3.1 Feature engineering

The raw categoricals had two recurring issues: high cardinality with sparse tails, and income signal concentrated in a few meaningful distinctions rather than spread across all categories. The fix was rule-based binning that preserved the income gradients seen in EDA while collapsing the tail. Education went from 20 categories to 6 bins (no education, high school, some college or associates, bachelors, masters, graduate or doctorate). Marital status went from 7 to 3 (married, never married, previously married). Country of birth was grouped into 6 regions. Occupation and industry were each collapsed into 4 income-tiered bins.

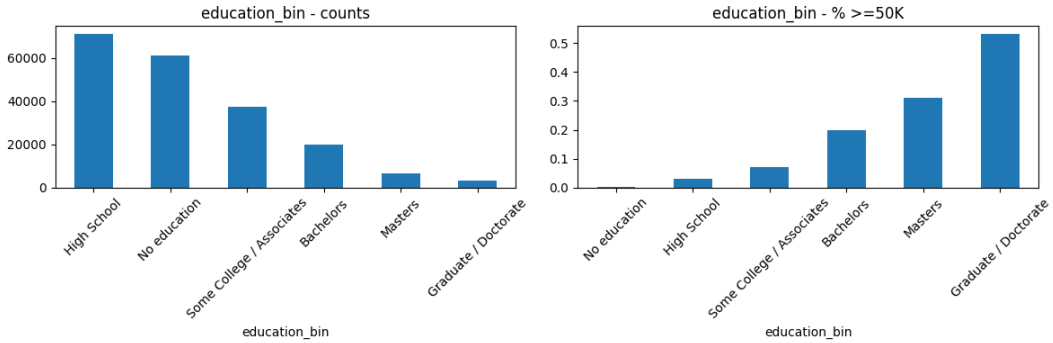

Figure 2: High-income rate by education bin after binning 20 raw categories into 6. The income gradient is preserved cleanly, with a near-monotonic rise from no education through to graduate or doctorate. This is the kind of structure the binning was designed to keep.

Financial features were transformed with $\log(1+x)$ and paired with binary presence flags. A single `education_employment` interaction term was added because the EDA showed graduate or doctorate full-time workers reached above 60% high-income rate, which is much higher than what either feature predicts on its own. A linear model cannot pick that up without an explicit interaction.

The survey-year variable was excluded from training. The dataset spans two adjacent years and there is no business reason for the model to learn that 1994 records behave differently from 1995 ones.

After all the engineering, the final feature set was 6 binary flags, 4 numeric features (including the log-transformed financial signal and weeks worked in year), 16 binned categorical features, and 15 raw categoricals kept at their original granularity for variables where binning would have lost signal.

4 Modeling approach

I evaluated two model families. Logistic regression as the interpretable baseline, with `class_weight='balanced'` to address the imbalance. It trains in seconds, needs no tuning, and sets a floor: if XGBoost cannot clearly beat it, the added complexity is not worth it.

XGBoost is the primary candidate. It handles mixed feature types without distributional assumptions, supports the `scale_pos_weight` parameter for native imbalance handling, captures non-linear interactions like the education-employment combination, and produces feature-level attributions usable for business interpretation.

I trained XGBoost in three stages. A baseline with the default parameters (`n_estimators=300, max_depth=6, learning_rate=0.1`) and `scale_pos_weight` set to the negative-to-positive ratio. A tuned version using `RandomizedSearchCV` with 50 iterations of 3-fold CV, scoring on average precision, over learning rate, max depth, subsample, column sampling, gamma, and L1 regularization. Then a calibration pass using `CalibratedClassifierCV` with isotonic regression and 5-fold internal CV [2]. Calibration matters because raw XGBoost probabilities are known to be miscalibrated under class imbalance, and the client may want to set a probability threshold instead of a rank cut.

All models were evaluated on a stratified 80/20 train-test split. The final evaluation also reports metrics weighted by the census sampling weight, because the unweighted test set over-represents some strata relative to the general population. Bootstrap confidence intervals (200 resamples) were computed for every reported metric so the comparisons across models are not just point estimates.

5 Classification results

Table 1 shows the five model variants on the held-out test set. The calibrated, weight-evaluated XGBoost is the best across every metric.

Model	PR-AUC	Brier	P@10%	R@10%	R@5%
Logistic regression	0.547	0.045	0.361	0.707	0.510
XGBoost base	0.638	0.043	0.388	0.738	0.546
XGBoost tuned	0.640	0.043	0.388	0.739	0.547
XGBoost tuned + calibrated	0.642	0.038	0.389	0.741	0.549
XGBoost calibrated, weighted eval	**0.655**	**0.035**	**0.395**	**0.751**	**0.558**

Table 1: Test-set metrics for each model variant. P@k and R@k are precision and recall when the top k percent of scored customers are mailed. The weighted row applies census sampling weights at evaluation time, not training.

A few things in this table are worth considering. Hyperparameter tuning contributed essentially nothing: the base and tuned XGBoost rows have overlapping bootstrap CIs at every cutoff. The work was done by calibration and by weighted evaluation. The logistic regression gap, by contrast, is statistically significant: at the 10% mailer budget, the calibrated weighted XGBoost captures 75.1% of true high earners versus 70.7% for logistic regression, with non-overlapping CIs. On the test set that is roughly 600 additional correctly identified high-income customers at the same spend.

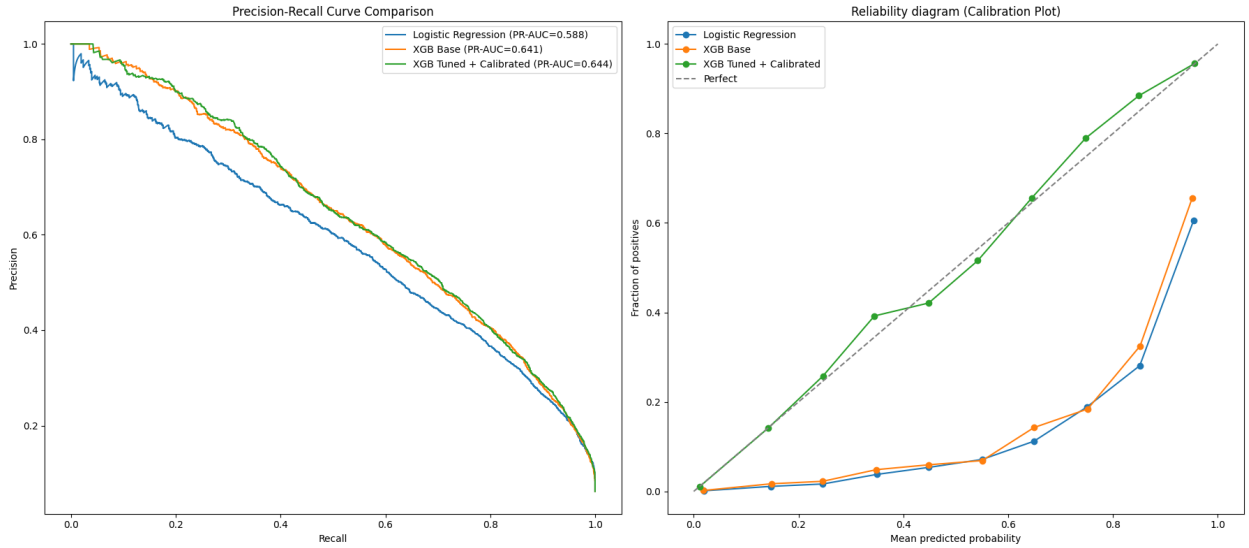

Figure 3: Precision-recall curves and reliability diagram for the three main model variants. Left: XGBoost dominates logistic regression across the full recall range, not just at extreme thresholds. The base and calibrated XGBoost curves are nearly identical, confirming that calibration did not hurt ranking performance. Right: the calibrated model tracks the diagonal closely below 0.5, but overshoots above 0.6, meaning it slightly overestimates the true positive rate in the high-confidence range. A user setting a high probability threshold should know the scores are mildly optimistic there.

The PR curve in Figure 3 shows XGBoost dominating across the full recall range, not only at extreme thresholds, which rules out the gap being a threshold artifact. The calibration plot confirms the model is well-behaved in the 0 to 0.5 range where most observations sit, with mild overconfidence above 0.6.

5.1 Feature attribution

SHAP values [3] were computed on the tuned XGBoost (Figure 4). Calibration only remaps probabilities and does not change tree splits, so attributions are identical to the calibrated model. The top features confirm the EDA hypothesis. Weeks worked in year and tax-filer status are the two strongest signals, both with the expected sign. Age 0 to 20 being the active bin is the third strongest feature and pushes scores down, correctly encoding that children never earn above $50K.

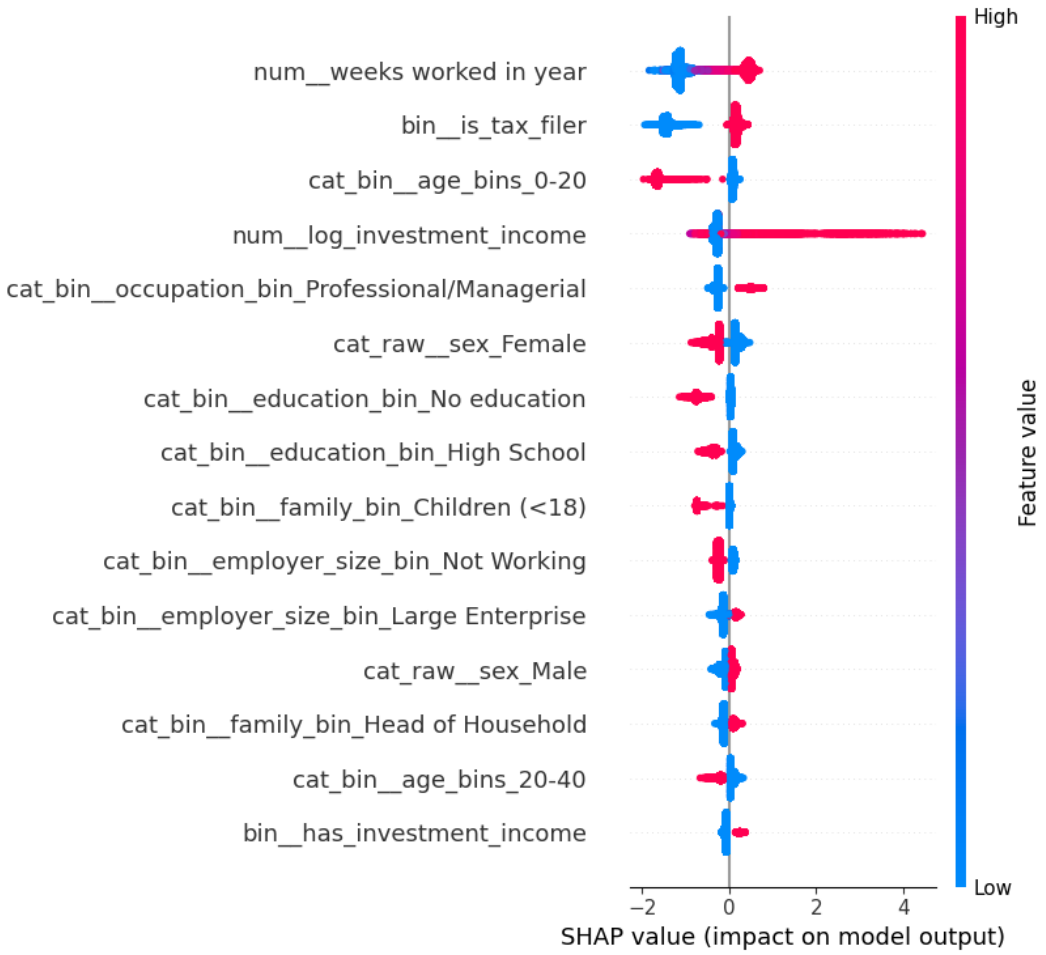

Figure 4: SHAP summary plot for the top 15 features. Each dot is one observation; horizontal position is the feature's contribution to the model's log-odds output, color is the feature value. Log investment income has the widest horizontal spread of any feature: most observations sit near zero with little impact, but a small number of high-investment-income individuals get a large positive boost. The female indicator pushes scores negative consistently, a direct reflection of the 1994 wage gap embedded in the data.

6 Segmentation

For the segmentation task I reused the classification preprocessor to get a consistent feature representation, then applied TruncatedSVD [8] with 30 components to reduce dimensionality before clustering. Thirty components retained enough variance for the clustering to be stable while making Gaussian Mixture Model fitting tractable on 200,000 rows.

I tested k from 3 to 7 using silhouette score on a 10,000-row sample. Silhouette peaked at $k = 3$, but I picked $k = 4$. The reason is purely operational: $k = 3$ collapses the immigrant-household segment into either the working-adult segment or the non-participant segment, and that distinction matters for marketing because product fit and outreach approach differ. *Picking the higher-business-value k over the higher-silhouette k is a deliberate tradeoff, not an oversight.*

6.1 Segment profiles

Table 2 summarizes the four segments. Cluster numbers are arbitrary GMM labels and would change on a reseed.

Segment	% pop	Weeks worked	Tax filer	High-income rate	Defining feature
Non-participants (Cluster 1)	39.4%	1.4	28%	1%	No labor force engagement; includes children
Core working adults (Cluster 2)	32.5%	47.0	97%	13%	Full-year work, married, professional or managerial
Immigrant households (Cluster 0)	18.1%	27.0	—	7%	60% native, 31% both-parents-US (population avg is much higher)
Younger transitional (Cluster 3)	10.0%	24.5	62%	5%	Never-married dominant, lower across all labor indicators

Table 2: Four-segment profile from the GMM. The high-income rate is computed post hoc using the classification target as a validation signal, not as a clustering input.

The cluster identities are easiest to read off the z-score heatmap in Figure 5, which shows how each cluster deviates from the population mean on the most informative features.

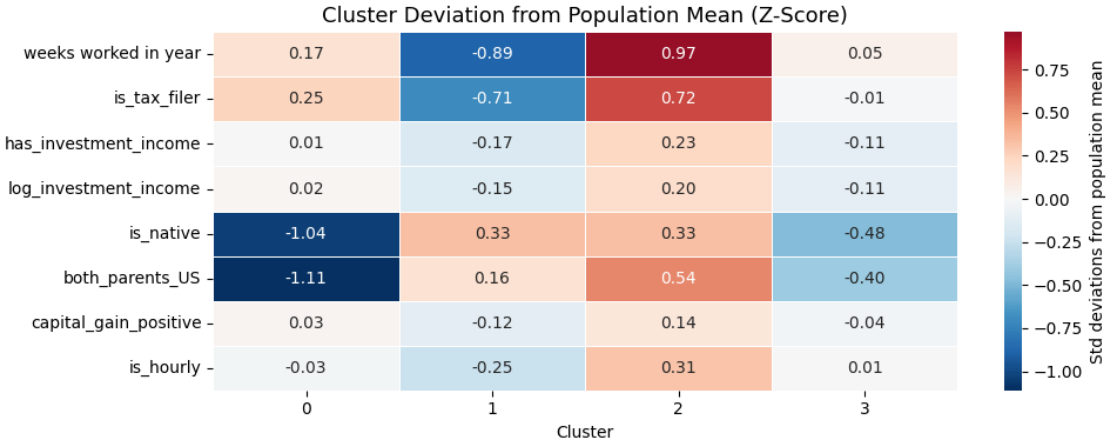

Figure 5: Cluster deviation from the population mean, in standard deviations, for the eight most informative features. Cluster 2 is positive on weeks worked and tax filer, the two features the classifier identified as strongest. Cluster 1 is the mirror image. Cluster 0's identity is entirely in the nativity rows: `is_native` at -1.04 and `both_parents_US` at -1.11 are the darkest cells in the heatmap, with everything else near zero. Cluster 3 is a weaker version of the same nativity pattern combined with mild below-average labor engagement.

6.2 Stability

Cluster assignments were re-fit on a bootstrap resample of the training data with a different random seed. The Adjusted Rand Index [5] between the original and bootstrap labels was 0.585, which is moderate stability, not high. Cluster 2, the prime marketing segment, was the most consistent

across bootstrap runs, while the boundary between Cluster 0 and Cluster 3 was the source of most of the instability. For deployment, that means Cluster 2 is the operationally trustworthy segment; the others should be re-validated periodically.

6.3 Marketing recommendations

Segment	Action	Product fit
Non-participants	Exclude from all targeted campaigns	None; mailers waste budget
Core working adults	Highest-priority targeting	Premium products: wealth management, high-limit cards, mortgages
Immigrant households	Standard outreach with multilingual materials	Retail banking, credit-building products
Younger transitional	Brand loyalty over premium products	No-fee checking, starter cards; convert as they mature into the core segment

7 Limitations and future work

The dataset is from 1994 and 1995. Patterns involving sex, race, immigration, and country of origin reflect the structural inequalities and demographic composition of that period and should not be deployed against modern customers without revalidation on current data. The gender gap in particular is pronounced in the SHAP attributions, and using it directly to drive marketing decisions would be both ethically and legally questionable.

Segmentation was run on unweighted training rows, so the segment-size percentages describe the sample, not the general population. For a marketing use case where the client cares about how many real customers fall into each segment, the GMM would need to be re-fit with sample weights or the assignments re-weighted at scoring time.

The calibrated model overshoots above 0.6 predicted probability. Any deployment using a fixed high-confidence threshold should account for this, either by recalibrating in that range or by sticking to rank-cut deployment.

A few things were out of scope for time. LightGBM and CatBoost were not tested, and a stacking ensemble of logistic regression and XGBoost might have closed the small remaining calibration gap. The survey-year column was excluded from features but not used as a held-out validation split, which would have given a more honest read on temporal generalization.

A Appendix

A.1 Full results table at multiple thresholds

Model	P@5%	R@5%	P@10%	R@10%	P@20%	R@20%
Logistic regression	0.604	0.487	0.437	0.704	0.276	0.888
XGBoost base	0.647	0.521	0.465	0.750	0.280	0.903
XGBoost tuned	0.645	0.520	0.465	0.749	0.282	0.908
XGBoost tuned + calibrated	0.649	0.523	0.464	0.748	0.283	0.913
XGBoost calibrated, weighted	**0.657**	**0.526**	**0.473**	**0.753**	**0.292**	**0.917**

Table 3: Precision and recall at three campaign-size thresholds. The recall@20% column shows that a wider campaign captures over 91% of all high-income individuals, which is the right operating point if reach matters more than precision.

References

[1] Saito, T. and Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE* 10(3).

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